

An Exploration Dynamical Systems Chaos Argyris C01 Completely Re-
vised and Enlarged
Second Edition
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Preface to the Second Edition

In 1994, when the first edition has been published in English and in German, we did not anticipate that this introduction to the field of non-linear dynamics and chaos would meet with such lively interest. Although attention to the subject has lessened in the media and in popular scientific publications over the last 20 years, the fundamental ideas, the theoretical insights and the tools developed for the analysis of non-linear dynamical systems have continued to spread into the most diverse areas of science and technology as well as into the respective literature and are nowadays part of the classical curriculum in many study programmes. Since the English edition sold out many years ago, we decided to follow up the expanded and revised second German edition with another extended edition in English.

After John Argyris passed away in 2004 at the advanced age of 91, we considered it our obligation to continue work on this book in the original sense – first in a second German edition – attempting to explain the complex topic of non-linear dynamics to a wide audience as descriptively and vividly as possible.

We took the opportunity to emphasise and further clarify the differences between purely temporal and spatio-temporal dynamics, in particular between chaos and turbulence. The initial hope – occasionally accompanied by euphoria – that insight into the special features of chaotic systems meant that scientists were about to solve the centuries-old problem of turbulence was not fulfilled. Without doubt, the elucidation of characteristics such as unpredictability and mixing in chaotic systems made a substantial contribution to understanding turbulent flows, yet the fundamental questions concerning fully developed turbulence are unsolved to this day.

By enlarging the circle of authors to include Rudolf Friedrich, we were able to add with Chapter 9 a completely new and more ambitious chapter

on turbulence, which comprises and critically scrutinises the fundamentals of turbulent flows and a series of classic turbulence models. While there is a multitude of excellent monographs on turbulence from the viewpoint of the engineering sciences, the new chapter mainly discusses the fundamental questions from the physical point of view, highlighting the common ground with concepts existing in chaos theory. To describe and investigate fully developed turbulence, it is indispensable to include probabilistic and stochastic methods. So that the reader can understand the chapter on turbulence better, we thought it appropriate to include two new sections in the mathematical introduction in Chapter 3: basic concepts of probability theory and invariant measure and ergodic orbits. Our aim in these sections was to present the basic ideas and concepts; we did not, however, strive for a complete description, rather referring the reader to the excellent literature on probability and stochastic processes.

In section 3.8.5, we added a brief introduction to wavelet transformation and outlined its relevance for time-frequency analyses of time series. The use of wavelets is especially favourable for analysing multifractal structures as described in section 8.5.2. In section 3.9.12, we also included a brief description of Markov analysis, a recently developed and far-reaching method which allows us to separate the deterministic part of the dynamics from the dynamical and even measurement noise. In section 9.6.8, this method is applied to velocity increments measured in turbulent flows. Section 8.7 contains an updated and extended section on routes out of chaos. Shilnikov bifurcations and spiral chaos received much attention in various applications, from chemical reactions to the propagation of nerve impulses and epileptic seizures. To understand such systems better, we also included a short overview in section 10.6. In addition, the paragraph on the kinetics of chemical reactions on surfaces has been largely revised in the hope of rendering this widely used field of application more comprehensible to a larger audience of students, see section 10.9. On the other hand, we dispensed with the section on celestial mechanics; this would have gone beyond the scope of this monograph.

Suddenly and unexpectedly, on 16th August 2012, Rudolf Friedrich passed away, taken from the midst of his life and work. His broad knowledge and interdisciplinary expertise made him extraordinarily inventive, not only in theoretical physics, his specialist field, but also in many other areas of science, including experimental and technical applications. We hope we have

been able to complete the work on this book as he would have wished.

Without the vigorous support of numerous friends and colleagues from many scientific disciplines, we would not have been able to finish this extended and updated second edition. To implement the original KeTeXversion in LaTeX and to generate and reconstruct the figures with the programme system AnT 4.669, we received professional help. We acknowledge the valuable hints for additions and improvements which we obtained in the course of many discussions and during proof-reading. We wish to thank all those who constantly and sedulously helped us and are especially indebted to Viktor Avrutin, Inna Avrutina, Rolf Bader, Anton Daitche, Martin Dziobek, Markus Eiswirth, Jan Friedrich, Jason Gallas, Svetlana Gurevich, Andreas Haase, Marion Hackenberg, Marcus Hauser, Oliver Kamps, David Kleinhans, Yuri A. Kuznetsov, Bernd Lehle, Pedro Lind, Johannes L^ulff, Joachim Peinke, Peter Plath, G^unter Radons, Michael Schanz, Susanne Schmidt, Daniel Stellbrink, Robert Stresing, Daniel H. Sugondo, Christian Uhl, Hans van den Berg, Judith Vogelsang, Georg Wackenhut and Michael Wilczek. Our special thanks go to Prudence Lawday, the translator of the first English edition, for her excellent help with this extended manuscript. We are also indebted to Michael Resch, the director of the High Performance Computing Center Stuttgart (HLRS) of the University of Stuttgart, for his generous assistance and to the Argyris Foundation for its financial support.

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Stuttgart, October 2014 M. Haase, G. Faust

1 Descriptive Synopsis of the Text

But thought's the slave of life, and life time's fool;
And time, that takes survey of all the world,
Must have a stop.

William Shakespeare, King Henry IV, Part I, IV, 2, 81

This book is conceived as an elementary introduction to the modern theory of nonlinear dynamical systems with particular emphasis on the explo-

ration of chaotic phenomena. One might ask why yet another book should be published when the literature on chaos and non-linear oscillations already fills shelf after shelf following the stormy developments in this branch of science since the 1970s. The reasons which prompted us have been detailed in the preface.

Our aim was to write an elementary, yet detailed text-book, without assuming an all-too profound mathematical knowledge on the part of the reader. It is aimed not at mathematicians and theoretical physicists, but rather at budding scientists such as chemists, biologists and engineers in all fields, particularly in the aeronautical and electrical engineering, but also in musical acoustics, as well as economic scientists and interested graduate students. Should they be required for a comprehension of the subject, mathematical procedures and tools are introduced in small sub-sections; in all cases, we have attempted to support the text illustratively by graphics, colour plates and additional computer demonstrations. A fairly extensive list of references is intended to help the reader in his quest for further reading.

Chapters 2 and 3 are an elementary introduction to the intellectual and conceptual edifice of non-linear dynamical systems. In Chapter 2, we present an outline of the complex of problems investigated in this book. We consider non-linear dynamical processes, restricting ourselves to deterministic and, with few exceptions, finite-dimensional systems which can be described by systems of ordinary differential equations. In contrast to classic treatises on the theory of oscillations which generally devote a great deal of space to the solution of linear problems and mainly consider special solutions of selected problems, the theory of dynamical systems founded by Henri Poincaré allows qualitative statements on the overall set of the solutions of specific as well as broad classes of differential equations by a combination of analytical and geometrical, differential topological methods. Central questions concern the structure of the phase space, the stability behaviour of the solutions, their longterm response and the effects of bifurcations, i.e. the qualitative changes in the flow as a result of variations in the parameters. The fact that even the simplest nonlinear systems can generate chaotic behaviour leads to the discussion in section 2.1 of the concept of causality and the connection between determinism and predictability. Subsequently, using the example of single-degree-of- freedom oscillators with and without friction, with and without excitation force, we describe possible states of motion of dynamical

systems – such as stationary, periodic and chaotic behaviour – and introduce some elementary basic concepts such as phase space, trajectories, manifold, attractors etc.

A central theme in the analysis of non-linear systems is the discussion of the stability characteristics of states of equilibrium, periodic motions etc. Chapter 3 thus begins with a discussion of the singularities of linear differential equations and a classification of the fixed points on the basis of eigenvalues and eigenvectors for the two-dimensional case as well as the possibilities of an extension of the results to non-linear systems. The introduction of a series of stability concepts is followed by the definition of the invariant manifolds which enable a structuring of the phase space. We then discuss Poincaré sections and discrete maps, followed by a first elementary meeting with the logistic map where we can observe a transition from stationary to periodic, multiple periodic behaviour to chaos through variations of a control parameter. General periodic motions can be easily described in Fourier space. Therefore, we continue with a brief introduction to harmonic analysis, including a short survey of the most important properties of the Fourier transform and a reference to wavelet analysis, which allows us to perform time-frequency analyses of non-stationary processes.

The subsequent section 3.9 presents a summary of the basic principles of the theory of probability. Although this book is devoted to the analysis of deterministic nonlinear systems, one often needs a set of fundamental terms and definitions stemming from probability theory to characterise the dynamics of chaotic systems in phase space and to describe turbulence. Among these are concepts such as random variables, probability density, conditional and joint probability and moments of distribution functions. We briefly outline the importance of the central limit theorem and present a short description of the most commonly used density distributions. Finally, we give a short overview of the recently developed Markov analysis, which allows for sequentially measured experimental data to disentangle the underlying deterministic dynamics from the often unavoidable superimposed fluctuating forces and to extract the basic stochastic equations. On the basis of probability theory, we introduce in the final section 3.10 the concepts of invariant measure and ergodicity.

Dynamical systems can be assigned to two basic families, namely conser-

vative systems not exposed to loss of energy and dissipative systems subject to loss of energy. In Chapter 4, we consider the conservative systems and attempt to span the gap between integrability and chaos. Within the concept of classical mechanics, the Hamilton equations are derived, taking d'Alembert's principle as a starting point. The theorem of Liouville allows us to investigate the volume conservation of the phase flow of conservative systems and thus to distinguish between conservative and dissipative systems. An analytical solution of non-linear Hamilton differential equations is only possible in exceptional cases, namely if the system can be transformed to action and angle variables and thus described by quasi-periodic motions on a torus in phase space. Such an integration is demonstrated on the example of Kepler's motion of a planet around the sun. In contrast, an analytical solution of the three-body problem is no longer possible. Towards the end of the 19th century, Poincaré succeeded in proving that small perturbations of the two-body problem throw doubt on the stability of the solutions. He showed that perturbation methods do not necessarily lead to correct results since the series used in the course of the calculation might diverge, thus possibly leading to false conclusions. It was not until 60 years later that, within the framework of the KAM theory, Andrei N. Kolmogorov, Vladimir I. Arnold and Jürgen Moser were able – under certain conditions – to answer the question of the stability of perturbed Hamilton systems. In section 4.4, the KAM theory is outlined and the disintegration of the tori in phase space demonstrated on Poincaré sections of two-dimensional perturbed Hamilton systems. It turns out that the emergence of heteroclinic points generates the irregular, chaotic behaviour in the neighbourhood of hyperbolic points. A detailed discussion of the conservative Hénon map closes this chapter.

Chapter 5 deals with dissipative systems. In the classical mechanics of Galilei and Newton, loss of energy due to friction was considered a disturbing factor and was thus neglected. However, this idealisation can be applied in a good approximation to the orbits of the planets, but only rarely to motions on earth. The theory of dynamical systems shows that the introduction of dissipative terms does not make the problem more complex, but rather simplifies the long-range solution in many cases. The long-term dynamics of high-dimensional dissipative systems is often conditioned by some few fundamental modes; a contraction of the phase space volume takes place and attractors emerge which cannot be generated in conservative systems. Dissipative systems not subject to energy supply come to rest, they approach a

state of equilibrium or point attractor. If the system is fed with energy, then, in the simplest case, periodic or quasi-periodic behaviour can occur or, on the other hand, completely irregular behaviour which is described in phase space by a strange attractor, an extremely complex structure with zero volume. The typical characteristics of such strange attractors are summarised using the example of the Lorenz attractor, the structure of which is characterised fundamentally by the sensitive dependence of the trajectories on the smallest perturbations of the initial conditions. In the subsequent sections of Chapter 5, we review a series of mathematical tools – namely the power spectrum, autocorrelation, the Lyapunov exponents and the dimensions in phase space – which allow a quantitative characterisation of and differentiation between the individual attractor types.

We begin with two classic criteria, the power spectrum, which supplies information on which basic frequencies dominate, and autocorrelation, which reflects the memory, i.e. the temporal interrelations of a signal. Both criteria are based on Fourier transformation, the most important characteristics of which are summarised previously in section 3.8. While demonstrating a differentiation between regular and irregular behaviour, they are not sufficient to characterise chaotic motion. Further mathematical tools are necessary, such as the Lyapunov exponents, for example, which provide a criterion for the stability behaviour of neighbouring trajectories. The classic Floquet theory, which allows a linear stability analysis of periodic motions, is extended to irregular courses of motion and leads to the spectrum of the Lyapunov exponents. Section 5.4 closes with the description of algorithms which allow a numerical calculation of all Lyapunov exponents.

The concept of dimension represents a further important instrument for the classification of the various attractor types. Damping generally ensures that the number of independent variables determining the dimension of the phase space are reduced considerably during the transient phase. The dimension of an attractor is thus a suitable source of information on the number of fundamental modes participating in the long-term response: a point attractor of dimension 0 corresponds to a state of equilibrium, a limit cycle of dimension 1 to a periodic motion etc. A strange attractor, which, for example, is a sort of hybrid between a surface-like and a spatial structure in a three-dimensional phase space, corresponds to a chaotic motion. Classic dimension concepts are not adequate to describe its intricate, mille-feuillelike

structure; it has to be ascribed a non-integral dimension between 2 and 3.

The French mathematician Benoît Mandelbrot coined the word “fractals” for such sets. What is responsible for the complexity of fractal sets is their self-similarity, i.e. their invariance with respect to alterations in scale. There is a whole family of suitable dimension concepts – such as the capacity dimension, the information and correlation dimension etc. – which we shall discuss in detail in section 5.5.

Dimensional concepts are often used to characterise experimental time series. If a system is based on deterministic laws, the temporal evolution of a single variable contains all the information on the total dynamics of the system. We discuss a reconstruction technique which offers the opportunity of reconstructing attractors from one-dimensional time series and also applying in this context concepts such as dimension (section 5.5.4) and the Lyapunov exponents (see also section 10.9) to characterise the associated orbits. We conclude Chapter 5 with the discussion of the Kolmogorov-Sinai entropy, based on Shannon’s information theory and quantifying the degree of disorder of a system. With the example of a simple map, the Bernoulli shift, it is demonstrated that the KS entropy measures the information transport from a microscopic to a macroscopic scale. Moreover, it allows for the specification of a bound for the time-span within which reliable prognoses on the course of the trajectories are still possible.

Typically, one or more control parameters appear in the model equations describing physical systems. If these differential equations have non-linear character, a variation of the parameters can result in a qualitative change of the topological structure of the solutions. Such changes are referred to as bifurcations. Chapter 6 contains a detailed discussion of the methods of the local bifurcation theory; we concentrate on local bifurcations of states of equilibrium and periodic motions arising from the variation of one control parameter. As an introduction, we first present some typical bifurcations of fixed points and elucidate the qualitative change of the solutions with corresponding phase portraits.

In the neighbourhood of a hyperbolic fixed point, it suffices to investigate the linearised system in order to establish the stability and solution behaviour. At bifurcation points, however, a fixed point loses its hyperbolic character. In this case, non-linear terms must be included in our considera-

tions in order to clarify the stability and the solution behaviour. To this end, the method of the centre manifold is applied. This approximation technique, presented in section 6.2, leads to a drastic reduction of the number of equations necessary to describe the long-term behaviour in the neighbourhood of non-hyperbolic fixed points and hence the clarification of the stability.

It is the purpose of the local bifurcation theory to classify the local bifurcations of fixed points, limit cycles etc. and to set up their normal forms. In section 6.3, we describe the method of normal forms, a technique with which non-linear differential equations in the neighbourhood of fixed points and limit cycles can be reduced to the simplest form by a series of non-linear transformations; the non-linear terms are eliminated successively whenever possible. A complete linearisation is only possible if no resonances occur. Normal forms of bifurcations, however, necessarily contain resonances.

In section 6.4, we discuss the bifurcations of one-parametric flows, deduce the corresponding normal forms and list the conditions which allow a classification of the individual bifurcation patterns. We then consider the question of the robustness of the individual bifurcations, i.e. we investigate whether small alterations in the model equations generate qualitative changes of the bifurcation patterns. Finally, we discuss the normal forms of bifurcations of fixed points of one-parametric maps. Since periodic orbits appear as fixed points in the Poincaré map, the discussion also includes the bifurcations of limit cycles.

In section 6.7, we demonstrate with the example of the logistic map that a continuous increase of the control parameter can generate a series of local bifurcations, in this case a cascade of period doublings, ultimately leading to chaotic behaviour; this latter is interrupted repeatedly by windows of regular behaviour, however. Mitchell Feigenbaum was the first to prove that this route to chaos is not restricted to the special form of the logistic map, but rather is a universal scenario which can be observed in many systems. In this section, we explore the background to the mechanism of continued period doublings and explain how the self-similarity characteristics of the bifurcation diagram can be decoded with the aid of renormalisation techniques and how the Feigenbaum constants can be determined. We then turn to the link with second-order phase transitions, known to us, for example, from the transition from a liquid to a gaseous state or from the ferromagnetic transition.

We conclude Chapter 6 with an introductory outline of synergetics, a branch of science founded at the end of the 1960s by the physicist Hermann Haken which investigates the dynamics of pattern formation processes. The central question of synergetics is whether and to what extent the formation of structures in the most varied realms of physics, biology, sociology, medicine etc. in animate and inanimate nature is based on generally valid principles and how this self-organisation can be reproduced in a mathematical formalism. On the one hand, we refer to the features in common with the centre manifold theory; on the other, we draw attention to the more extensive framework of synergetics, a result of the inclusion of spatial inhomogeneities and fluctuating forces.

Chapter 7 is devoted to the structure formation in convection flows and the deduction and discussion of the Lorenz system, which stimulated the development of the chaos theory fundamentally. The chapter begins with a qualitative description of the physical laws leading to the formation of the characteristic convection rolls in the Rayleigh-Bénard experiment. There follows a detailed deduction of the basic hydrodynamic equations for convection problems. The continuity equations, the Navier-Stokes equations and the heat equation are deduced from the conservation theorems of mass, impulse and energy. In many cases, these basic equations can be simplified by means of the Boussinesq-Oberbeck approximation, in which the variation of the density is only accounted for in the buoyancy term.

In the early 1960s, the American meteorologist Edward N. Lorenz applied this system of basic equations to study the accuracy of weather forecasts. While his highly simplified model for the Bénard problem cannot stand up to an experimental verification describing the onset of turbulence, it nevertheless led to a new paradigm in the natural sciences: On the one hand, there are basic limits to the predictability of a dynamical system in the latent presence of chaotic behaviour; on the other hand, however, time series which have been classified as purely stochastic in the past have turned out to describe in certain cases chaotic motions, the essential definition of which can often be construed with a restricted number of non-linear equations.

In section 7.3, we follow the original publications of Saltzman and Lorenz and deduce the three ordinary differential equations of the Lorenz system.

Subsequently, the evolution of the solutions of this system is described for variations of the free parameter which controls the temperature difference between the upper and the lower plate in the Rayleigh-Bénard problem; the emergence of local and global bifurcations is observed. A series of colour plates demonstrates on the one hand the spatial and temporal evolution of the convection rolls in the Lorenz model and on the other hand the significant changes of the phase space structure due to an increase of the control parameter; these are illustrated by consideration of the evolution of the stable manifolds.

In Chapter 8, we present a series of mathematical models describing the transition from regular to chaotic motions. In the early years of chaos research, the hope was that these scenarios could describe the transition to turbulent behaviour, thus making a substantial contribution to the solution of the turbulence problem.

The first model describing the onset of turbulence goes back to Lev D. Landau (1944), who assumed that the transition from laminar to turbulent behaviour is generated by an endless series of (Hopf) bifurcations. It was not until a quarter of a century later that the deficiencies of this model were recognised: Landau's model cannot explain the sensitive dependence of the flow on small perturbations nor generate a mixing of the trajectories in the phase space. In 1971, David Ruelle and Floris Takens proved that generally quasi-periodic motions on higher-dimensional tori are not stable, as Landau had presumed, but rather that the quasi-periodic motion becomes chaotic abruptly after the emergence of the third incommensurable frequency, i.e. the motion on a T 3-torus collapses into a strange attractor, a designation coined by Ruelle and Takens. A great step forward had been taken and a window opened towards the description of the characteristics of turbulence – such as unpredictability of the motion and mixing – on the basis of deterministic equations. With the aid of refined measuring techniques, Gollub and Swinney succeeded in achieving an experimental verification for the Bénard problem based on the analysis of the power spectra and supported by the determination of the dimension of reconstructed attractors.

Sections 8.3 to 8.5 contain a discussion of the transition from quasi-periodic to chaotic motions using the circle map as a model. Two different scenarios are conceivable. Either chaos occurs following a synchronisation

of the frequencies and a subsequent series of period doublings – this is the Feigenbaum route already discussed in section 6.7 for the logistic map – or there is a direct transition from quasi-periodicity to chaos. The second route requires the tuning of two control parameters and can also be interpreted by means of the circle map. Both transitions to chaos are of universal character and the scaling characteristics can be decoded in both cases with the aid of the renormalisation theory. A selection of figures aims at facilitating access to this complex material. There follows a description of the work of Albert Libchaber who succeeded in confirming the theoretical prognoses quantitatively by experiments which were highly precise at that time. Section 8.6 contains a discussion of intermittent transitions which, following Yves Pomeau and Paul Manneville (1980), can be classified in three groups.

Having presented a series of scenarios describing transitions from regular to chaotic behaviour, we present in section 8.7 an overview of various strategies allowing the reverse, a conversion of chaotic motions to regular regions. Following the seminal paper of Edward Ott, Celso Grebogi and James A. Yorke from 1990, this issue became a field of non-linear dynamics which experienced a rapid development. This is because it was recognised in many areas like medicine, chemistry, laser technology and telecommunication, as well as in other fields of engineering science, that chaotic behaviour contains a high potential with regard to a possible control and regulation of chaos. To this end, one takes advantage of a characteristic feature of chaos, the sensitive dependence of the dynamics on small perturbations, by imposing minimal control impulses which stabilise the unstable orbits inherently contained in strange attractors. In section 8.7 we give a brief survey of the most effective strategies and provide information on a multitude of applications with the potential to lead to innovative developments.

In the beginning, it was hoped that chaos research would help uncover the basic mechanisms leading to fully developed turbulence. These high expectations have not yet been fulfilled. The reason is that chaos theory researchers initially confined themselves to a purely temporal description of the dynamics, which is not sufficient for an understanding of turbulent flow. It is rather the additional spatial dependence of all quantities which leads to complex interrelations and to an extremely high dimensionality of the corresponding phase space.

Chapter 9 presents an overview of the state of turbulence research from the present point of view, almost 40 years after the paradigm shift in the understanding of non-linear processes. On the one hand, we identify the issues in which chaos theory has made substantial contributions to an understanding of turbulent flow. On the other hand, it becomes evident that the spatio-temporal complexity and the extraordinarily high number of excited modes in fully developed turbulence makes the use of statistical methods unavoidable.

In the chapter on turbulence, the main focus is on various vortex models and the discussion of their influence on the dynamics of fluid flow. A detailed description of the characteristic scales, of the energy cascade and of Heisenberg's turbulence model follows. Particular emphasis is put on the derivation of the Kármán-Howarth equation and an exposition of the underlying assumptions. Subsequently, we describe the classical turbulence models of Kolmogorov from 1941 and 1962 as well as more recent multifractal models presented by Frisch and She-Leveque and discuss the strengths and weaknesses of the respective models. We also report on an application of the Markov analysis presented in section 3.9 to the velocity increments of a turbulent jet. The fact that the turbulent data can be described by a Markovian process in scale allows us to express the full n -point statistics by the 2-point transition probability densities and to reproduce the intermittent behaviour accurately. Finally, we add an overview of some open questions and possible issues relevant for future research.

In the final Chapter 10, we present some fields in which the application of the theory of dynamical systems is very promising. With the aid of computer experiments we discuss a variety of classical models displaying features which are characteristic of a multitude of dynamical systems.

For non-linear systems, analytical solutions can only be given in exceptional cases; in the case of chaotic behaviour, we generally have to rely completely on numerical investigations. This is the explanation for the fact that, as a result of efficient computers, the theory of dynamical systems – once the domain of the mathematical and physical faculties – is now receiving increasing notice in the most varied disciplines such as biology, chemistry, meteorology, electrical engineering and hydrodynamics, as well as in economic sciences, as demonstrated by the flood of publications from these fields.

The discovery of universal characteristics, i.e. of fundamental principles common to all non-linear systems of a whole class of problems, was without doubt a milestone in the more recent history of the theory of dynamical systems and justified the intensive study of seemingly over-simplified systems. The investigations were accelerated greatly by the swift development of efficient computers – indeed, they were only made possible through this explosive growth of computer power.

The first section of Chapter 10 offers an insight into bone remodelling processes. In order to achieve a durable anchoring of endoprostheses, models are being developed with the aim of describing bone retrogression or bone formation on the basis of dynamical systems. Should this succeed, computer simulations could speed up the development of more reliable types of endoprosthesis. In a first study, two very simple dynamical systems for the modelling of bone adaptation processes are discussed. Specially constructed recursion rules are particularly suitable for elucidating general characteristics of dynamical systems due to the low computational effort. In particular, the dissipative Hénon map was conceived as a model for a Poincaré map of a three-dimensional system. With the aid of this map, we demonstrate in section 10.2 that a mixing of the trajectories in the phase space results from a continuous repetition of stretching and folding processes. This mechanism is the key to the understanding of the exponential drifting apart of neighbouring trajectories in the phase space; it is illustrated with the example of the Lorenz attractor in a series of colour plates.

In section 10.3, we return to the Lorenz equations and discuss some further characteristics of the system. A number of figures demonstrate the emergence of cascades of period doublings within regular regions and the coexistence of attractors. The bifurcation diagram of the Poincaré sections is highly reminiscent of that of the logistic map, thus offering evidence of universal laws. Conclusions about the system behaviour can be drawn by calculating the Lyapunov exponents. The section ends with the determination of the capacity dimension of the Lorenz attractor according to Hunt and Sullivan's procedure.

In section 10.4, we turn to the van der Pol equation, originally set up to describe electrical circuits. The system possesses a non-linear damping term

with the characteristic that the system is supplied with energy for small amplitudes, which manifests itself in a negative damping term, whereas energy is dissipated for large amplitudes. Typically, self-excited oscillations occur in such systems, i.e. although no external periodic forces are applied, the system tends towards a limit cycle on which energy supply and dissipation are in balance in the temporal mean. We first consider the van der Pol equation without external excitation. In the case of a small damping parameter, i.e. weak non-linearity, we observe, as expected, almost sinusoidal oscillations; in the case of large damping values, on the other hand, we observe self-excited relaxation oscillations. Computer investigations suggest that exactly one limit cycle exists in the phase space; the strict mathematical proof of this observation with the aid of the Poincaré-Bendixson theorem is laborious. If small periodic excitation forces are applied in the case of low damping, the averaging method can be used; here, by taking a time average, the initial system is replaced by an autonomous system involving the neglect of higher harmonics. Large damping parameters respectively strong non-linearities together with periodic external excitation are discussed for a modification of the van der Pol equation. On the basis of Poincaré sections and phase portraits, the character of the chaotic solutions on the Birkhoff-Shaw attractor is illustrated.

The Duffing equation considered in the subsequent section represents one of the simplest dynamical systems in which chaotic behaviour can be observed. Differential equations of its type are obtained in the formulation of non-linear pendulum oscillations or for electrical circuits and other problems, for example. It approximates the behaviour of an externally excited non-linear oscillator with one degree of freedom. We deduce the Duffing equation for the example of an externally excited buckled beam. In this case, the equation of motion is a non-linear partial differential equation, the solution of which is approximated with the aid of the Galerkin procedure taking the first fundamental mode into account. As a result, we obtain the Duffing equation with negative stiffness. According to the selection of the control parameters in the Duffing equation, we observe a great variety of system response. The Duffing equation is integrated numerically for a series of parameter values and the results presented in the form of phase portraits and Poincaré sections.

A surprising phenomenon can emerge in dissipative systems when several

attractors coexist in the phase space. Even for regular behaviour, the boundaries between the basins of attraction can assume fractal character so that predictions on the longterm behaviour are no longer possible in this domain. With the aid of the Melnikov method, bounds for the control parameters for which fractal boundaries emerge can be calculated explicitly. For the example of the Duffing equation, the calculation of the so-called Holmes-Melnikov boundary is carried out explicitly.

In section 10.6, we briefly discuss some bifurcations of flow involving homoclinic orbits in three-dimensional vector fields. The most complex case, the so-called Shilnikov bifurcation, leads to a highly complex topological structure of the flow and can be observed, for example, in the propagation of nerve impulses along axons or during epileptical seizures.

By now, computer experiments play an important role in non-linear dynamics. In general, theorems cannot be proven with the aid of computers; nevertheless, numerical investigations in association with modern computer graphics reveal unexpected characteristics and links in non-linear systems which had previously not been taken into account. Seen in this way, numerical experiments are valuable aids nowadays. The two sections 10.7 and 10.8 illustrate the fact that graphic representations can be extraordinarily helpful for the comprehension of some phenomena occurring in non-linear recursion rules.

Shortly after the FirstWorldWar, the two French mathematicians Gaston Julia and Pierre Fatou studied the iteration of rational functions. They discovered that, depending on the initial value, the iteration sequences converged to different solutions; the boundaries between the basins of attraction of these attractors exhibit highly peculiar characteristics and are of unsuspected multifariousness. Due to the unavailability of computers, what was disclosed to the inner eye of Julia or Fatou remained hidden to the average scientist until, around 1980, Mandelbrot again turned to one of the simplest non-linear iteration rules in the complex plane, exploiting the opportunities offered by high-performance computers. Section 10.7 offers an insight into the manifold structures of Julia sets and the fractal boundaries of the Mandelbrot set. Calculation of the basin boundary of the point attractor $z = \infty$ for constant values of the control parameter of the mapping rule yield, depending on the parameter value, completely differing limit sets, the so-called

Julia sets. What is fascinating about Mandelbrot's recursion formula is that, in spite of its simplicity, such structural diversity emerges. A series of computer graphics shows connected fractal sets, dendritic forms as well as burst sets. The Mandelbrot set, on the other hand, encompasses all values of the complex control parameter for which the Julia sets are connected. A series of colour plates presenting various selective enlargements depicts the richness of structures in the fractal boundaries of the Mandelbrot set. In section 10.8, we again turn to the circle map of Chapter 8 and, by means of the Lyapunov exponents, gain an overview of the inner structure of the Arnold tongues. The results agree very well with the theoretical results. On the basis of the Lyapunov exponents, the different transitions to chaos can be discussed very simply.

Their self-similar structure is reflected in the corresponding bifurcation diagrams. In the last section of Chapter 10, section 10.9, we decided to present an overview of some interesting dynamic manifestations from the field of physical chemistry which display chaotic outbursts. We concentrate on the dynamic phenomena arising in the oxidation of CO on the surface of platinum crystals. Significantly enough, detailed experiments have been performed under isothermal low-pressure conditions. Extensive theoretical and experimental research work of high originality and distinction has been performed on this subject at the Fritz Haber Institute of the Max-Planck-Gesellschaft in Berlin under the direction of Gerhard Ertl.

In these extensive and varied numerical investigations and parallel experiments, a broad spectrum of complex temporal behaviour was observed, such as mixed-mode oscillations, a transition via period doublings to chaos and even hyperchaos, characterised by at least two positive Lyapunov exponents.

All these phenomena represent purely temporal evolutions as they would arise, for example, in a well-stirred medium; note, however, that in a heterogeneous reaction on a surface, stirring is not feasible. As a result, in heterogeneous reactions, more complex phenomena involving spatial dependencies will arise and complicate the model and the numerical analysis of such reactions considerably.

In a survey on CO oxidation on platinum crystals, Georg Ertl also discussed extensive experimental studies on spatio-temporal phenomena ob-

served by him and his collaborators at the aforementioned Fritz Haber Institute. The formation of stationary, propagating as well as soliton waves and even spiral chemical waves was recorded. Irregular behaviour, such as that of chemical turbulence, also emerged.

All the numerical investigations of the Fritz Haber Institute are based on kinetic models evolved for the description of the temporal dynamics on a platinum (110) single crystal surface. The rate constants and parameters arising in the equations have been studied extensively using a variety of surface science techniques in conjunction with complex detection and recording devices. We mention only parenthetically the application of low-energy electronic diffraction (LEED) studies of oscillations on platinum crystals, based on the motion of slow electrons; here, the researchers also observed periodic changes on the surface structure which oscillated between the $(1\text{\AA}\ 1)$ and the respective reconstructed $(1\text{\AA}\ 2)$ phase. Such structural surface changes in turn influence the evolution of other elementary processes and demand additional refinements of the model.

In view of the increasing complexity, the researchers at the Fritz Haber Institute refined the analytical model of oxidation stepwise, starting with the simplest simulation and then refining it step by step to account for additional effects and variables in order to describe the dynamical behaviour of increasing complexity of the CO oxidation on platinum (110) crystals. We elaborate on the importance of the individual steps since they clarify in an exemplary manner the stepwise development of models by illustrating the extent of validity and the limits of the respective models. We close this section with a short look at recent applications in biochemistry. Biochemical reactions involving enzymes which play the role of catalysts are, of course, much more multifarious and complex than the previously mentioned catalytical procedures. However, since biological systems are open systems exchanging energy and mass with their environment and since the occurring processes have a nonlinear character, it is once again indispensable to apply methods from non-linear dynamics and synergetics in order to take at least a small step towards understanding the complex interrelations of the system.